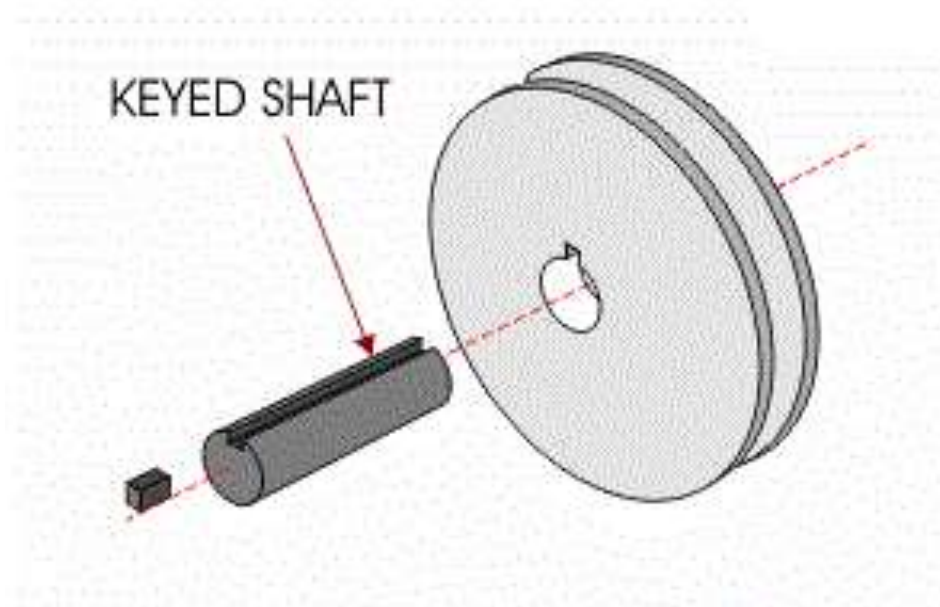


MECHANICAL POWER TRANSMISSION

TYPES OF MECHANICAL DRIVES

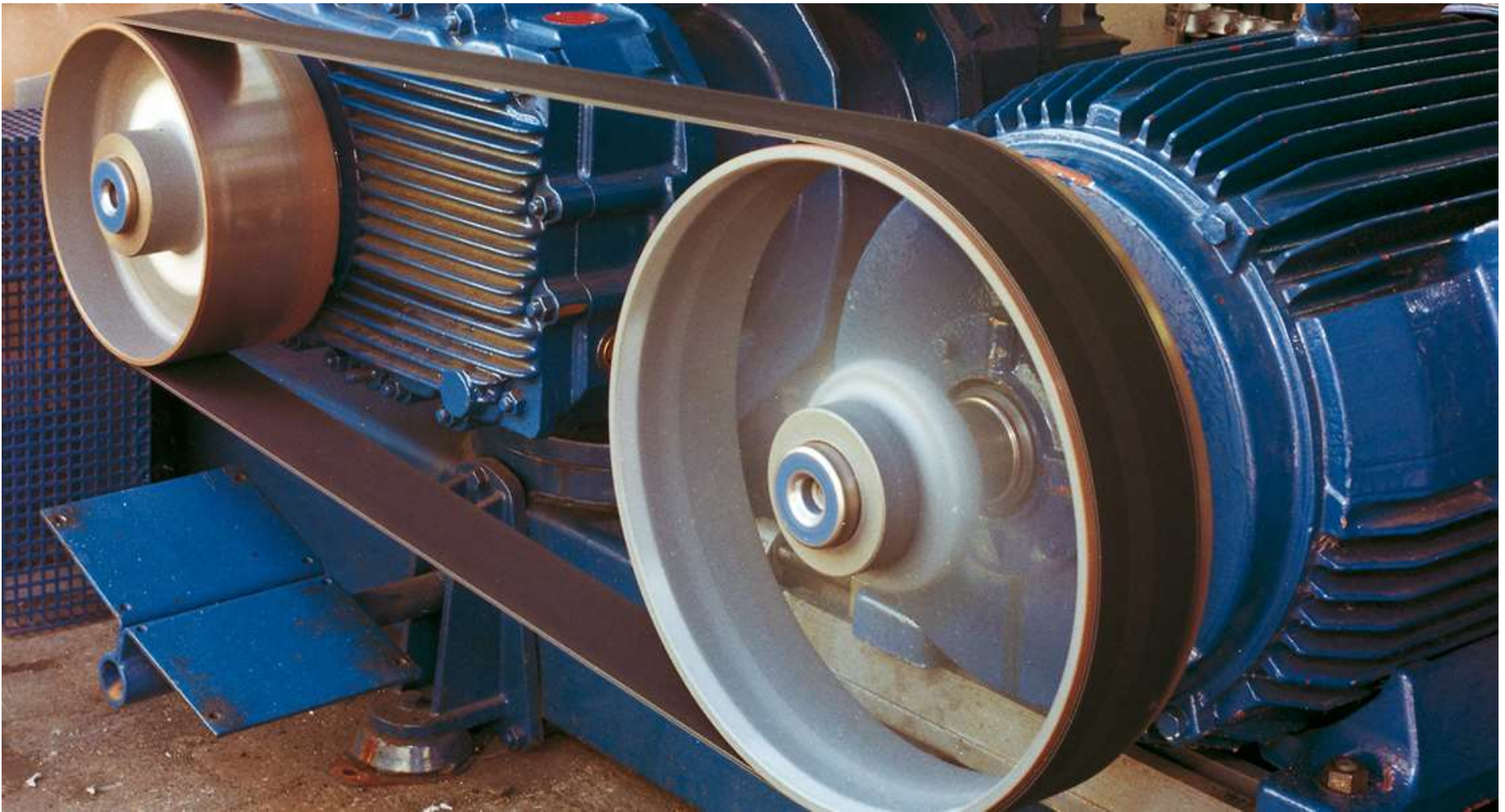
1. BELT DRIVE
2. ROPE DRIVE
3. CHAIN DRIVE
4. GEAR DRIVE

Shaft & Pulley



- Shafts: Driver Shaft, Driven Shaft
- Choice of drive depends upon,
 - ✓ Distance b/w shafts
 - ✓ Amount of power to be transmitted
 - ✓ Speed ratio of shafts
 - ✓ Accuracy

FLAT BELT DRIVE



V BELT DRIVE



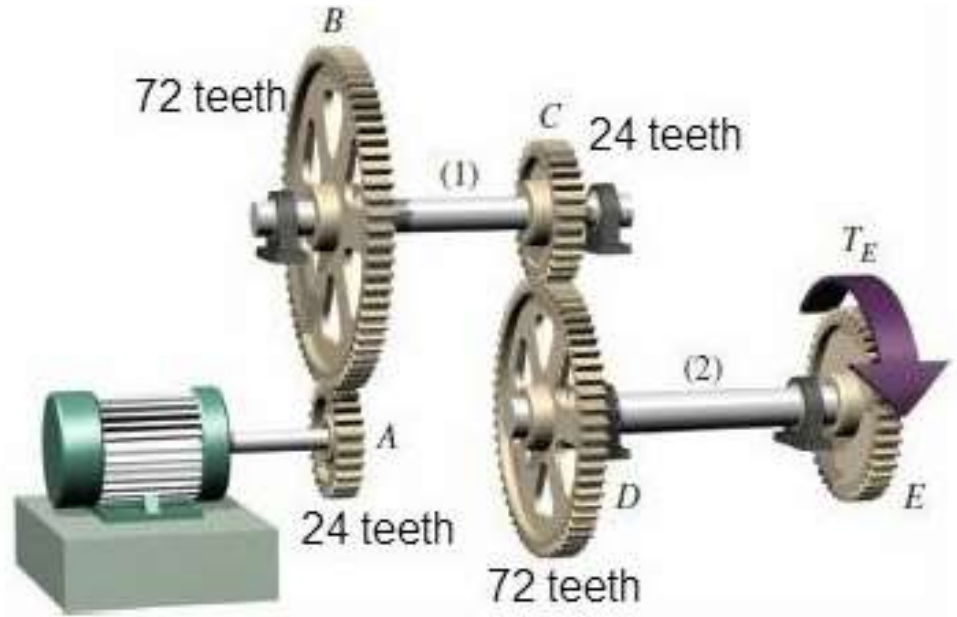
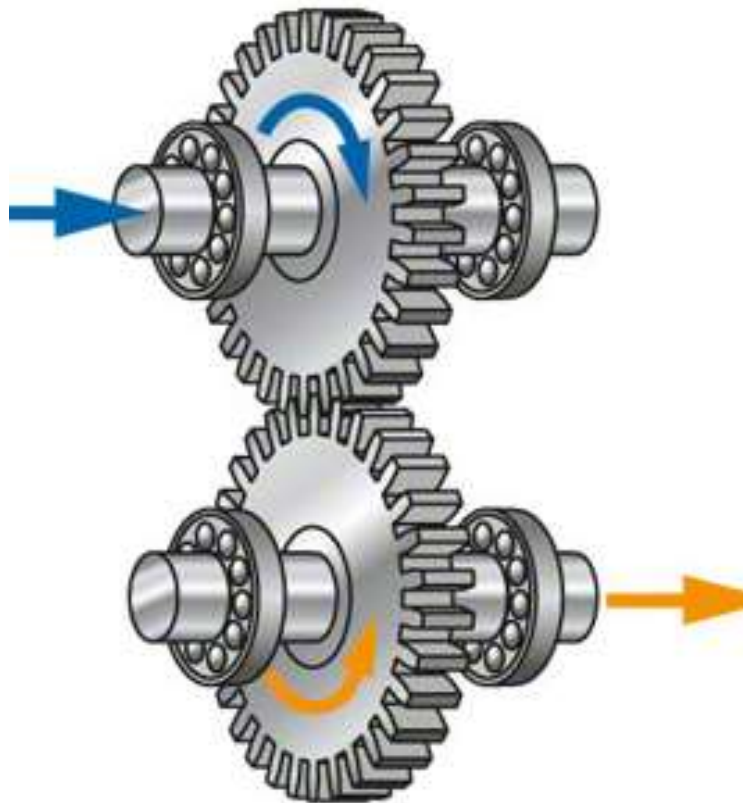
ROPE DRIVE



CHAIN DRIVE



GEAR DRIVE



BELT DRIVE

- Factor responsible for power transmission: **Frictional resistance** b/w belt and pulley.
- Amount of power transmitted depends on:
 - ✓ Velocity of belt
 - ✓ Tension
 - ✓ Arc of contact

Types of belts

Flat belts

- Moderate amount of power
- 10m distance b/w driving and driven shafts

V- belts

- Larger amount of power
- Shorter centre to centre distance
- Ideal Distance: 1.25 to 1.5 times dia of larger pulley

Properties of belts

- Large strength
- Flexibility
- Life
- High coefficient of friction

Commonly used materials

- Rubber
- Balata: 25% stronger than rubber
- Cotton: cheap, rough service
- Leather: Best pulling capacity, costly

Advantages of V-Belt

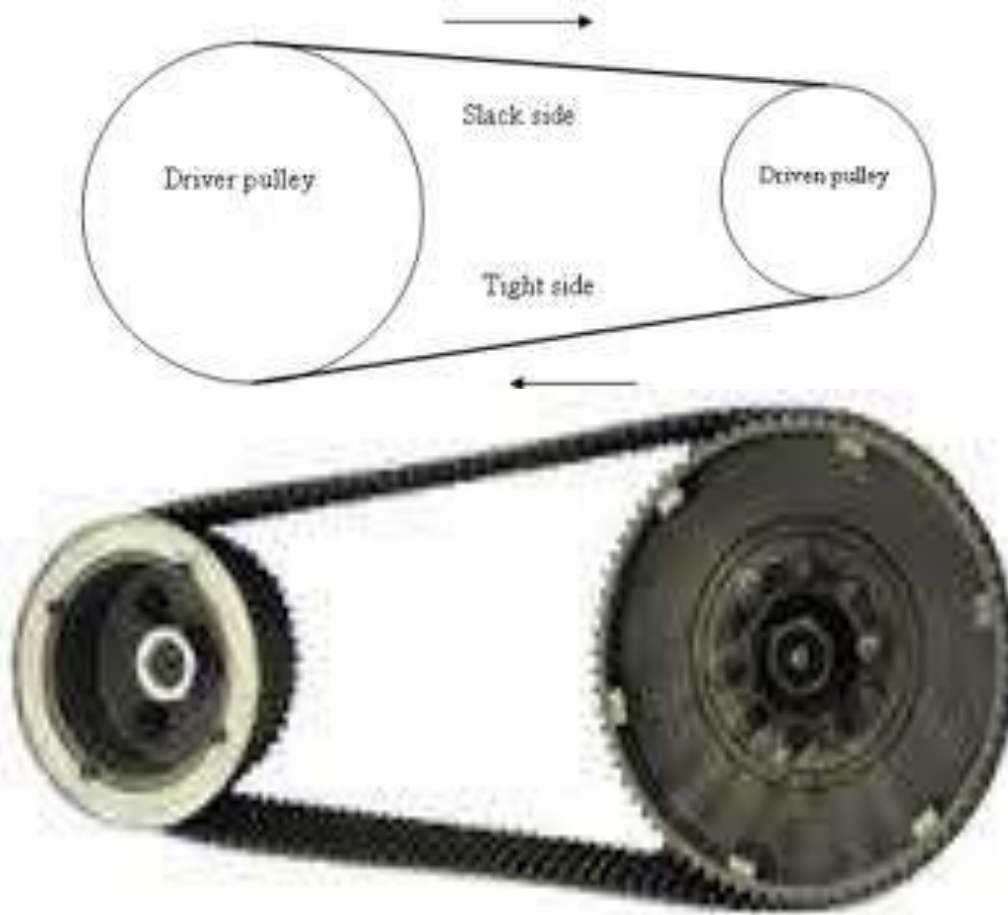
- No possibility of belt coming out of grooves
- Gripping action results in lower belt tension
- Can transmit more power
- Quiet operation



Types of belt drive

- Open belt drive
- Cross belt drive

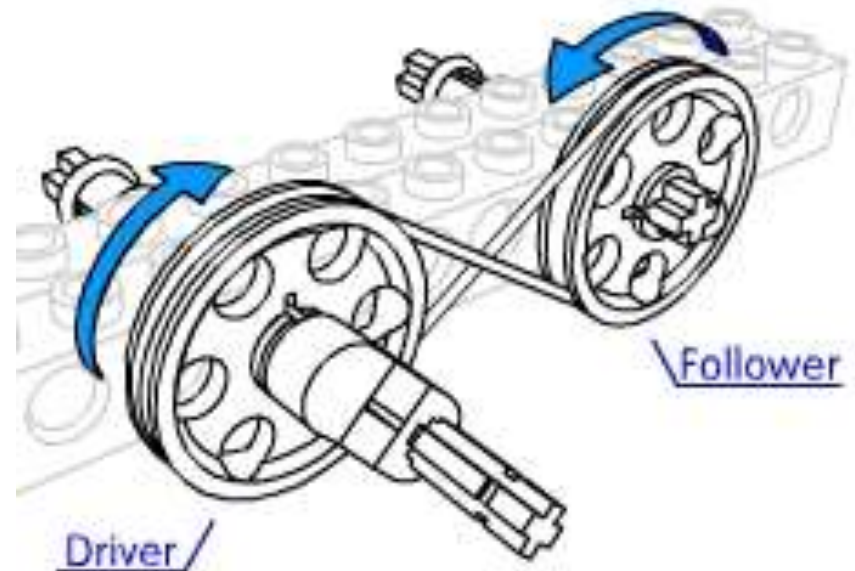
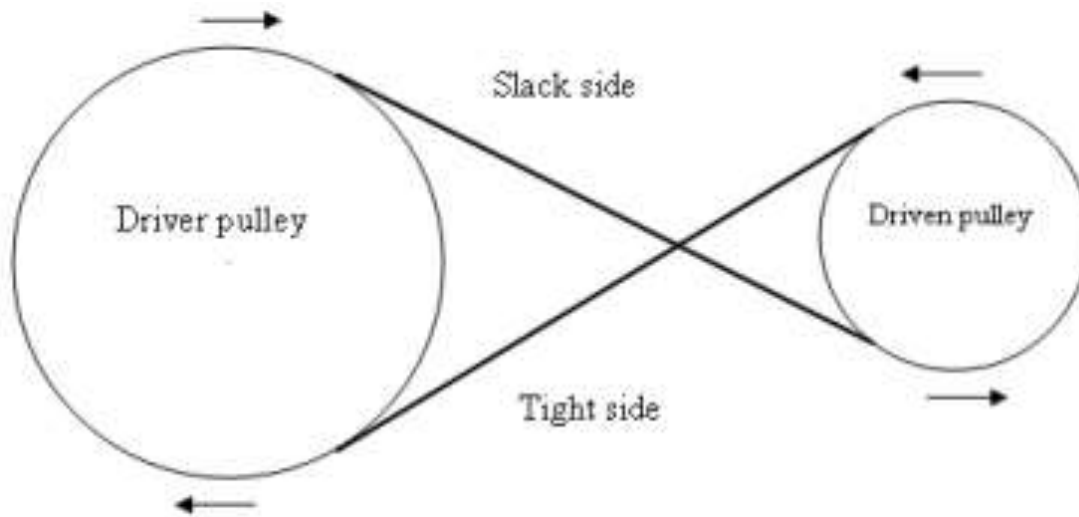
Open belt drive



Open belt drive

- Used when shafts are arranged in parallel and to be *rotated in same direction*
- *Driver pulley pulls* belt from one side and delivers to other side
- Tension on former side will be *greater than* later side
- Side where tension is more is called *tight side* and other side is called *slack side*

Cross belt drive



Cross belt drive

- Used with shafts arranged in parallel and to be *rotated in opposite direction*
- At point where belt crosses ,it rubs against each other and wear
- In order to *minimize wear* , shafts should be placed at a minimum distance of $20b$, b = width of belt
- Speed of belt should be less than 15m/s

Applications of belt drive

- Transmit power from low or medium capacity electric motors to operative machines
- Transmit power from small prime movers (I.C. engines to electric generators, agricultural & other machinery
- Conveyors

Advantages

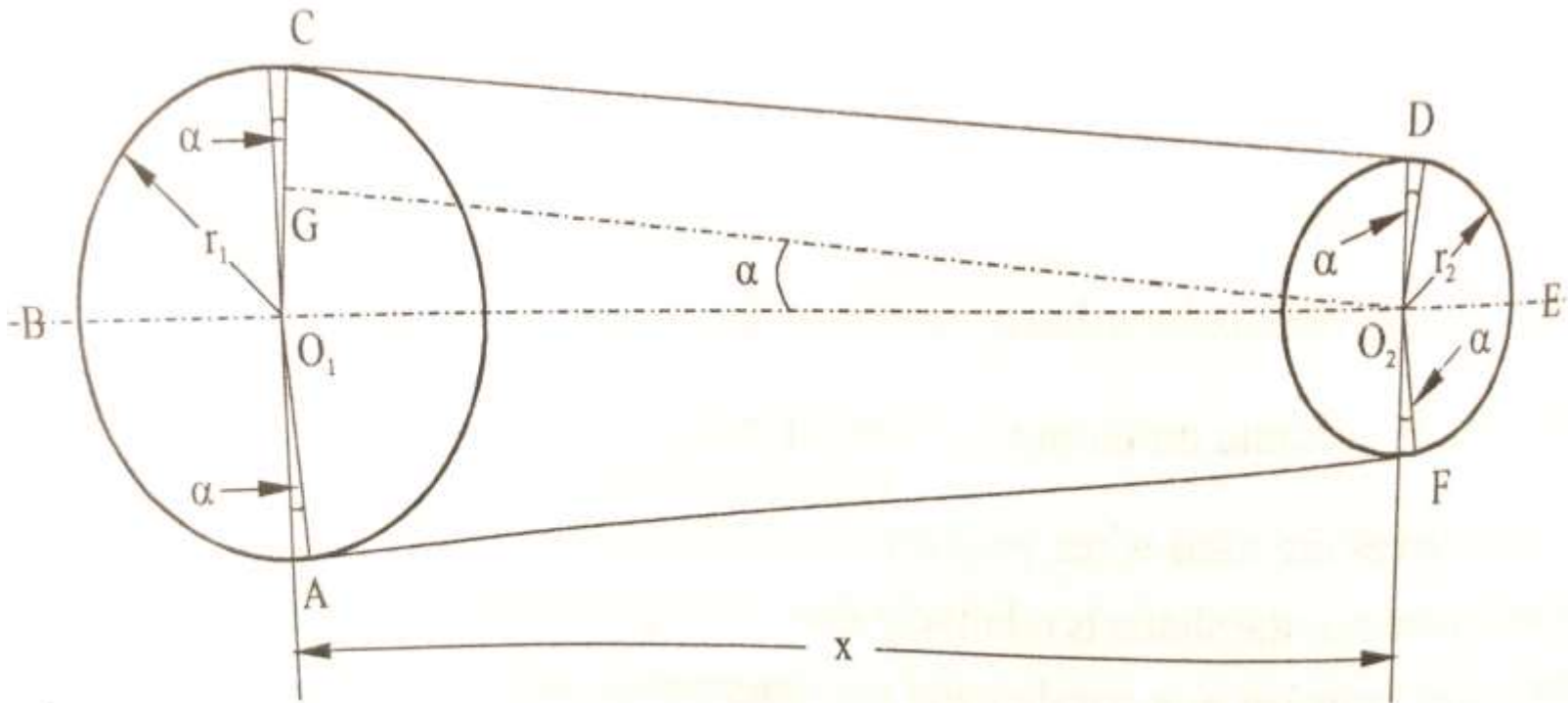
- Transmit power over medium distances.
- Smooth operation
- High speed operation
- Cost-relatively low

Disadvantages

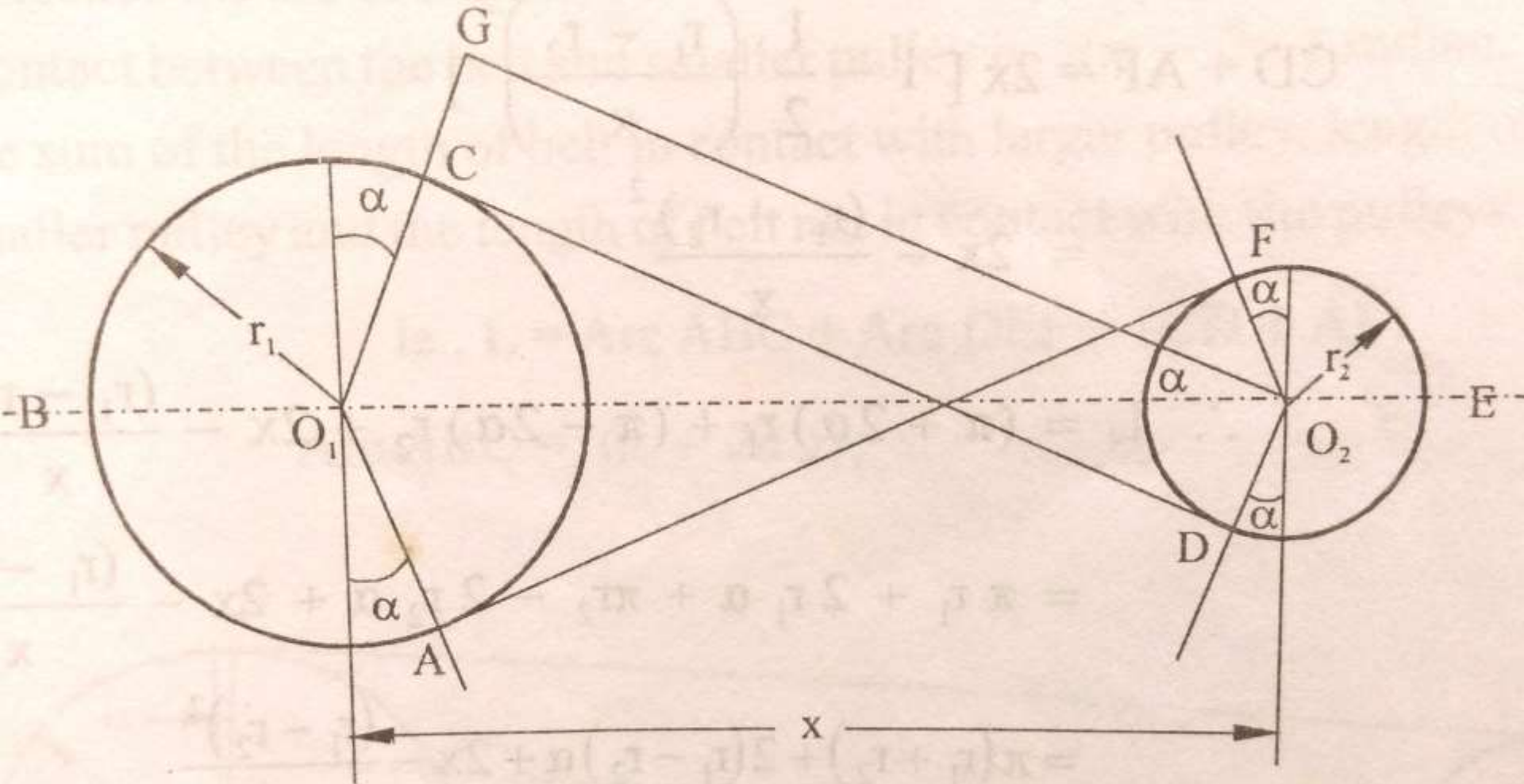
- Slipping of belt
- Belt tensioning
- Relatively short service life

Calculation of length of belt

1. Open Belt Drive



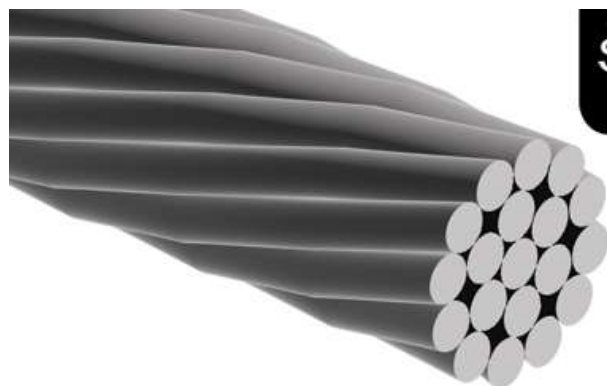
2. Cross Belt Drive



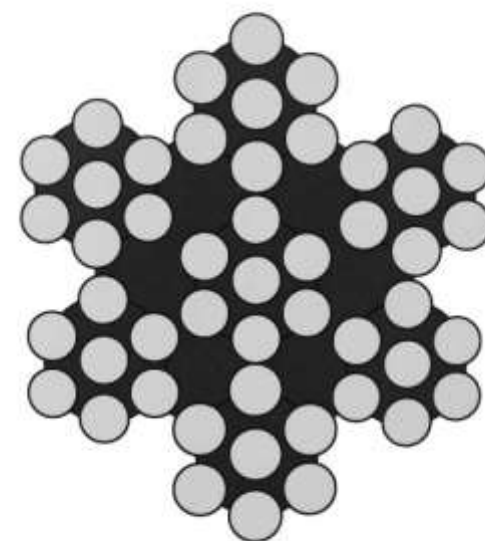
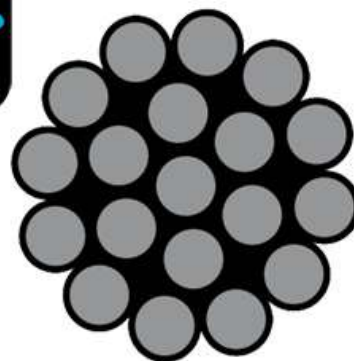
Rope Drive

- Usually made of cotton, having circular cross-section.
- Transmit considerable power over long distances.
- Ropes housed in grooved pulleys





SWR
1x19



- To **transmit large amount of power**, wire rope is used.
- It is made of cold drawn **wires**.
- Wires are twisted together to form a **strand**.
- A number of such strands are twisted together to form the **rope**.
- This reduces the bending stress on the rope.
- Ropes designated by no. of strands and no. of wires.
- For eg: (6 x 19) rope

Types of ropes

1. **Regular lay ropes**
2. **Long lay ropes**

Materials used

- Wrought iron
- Cast steel
- Aluminium alloys, copper bronze alloys
- Stainless steel

Applications

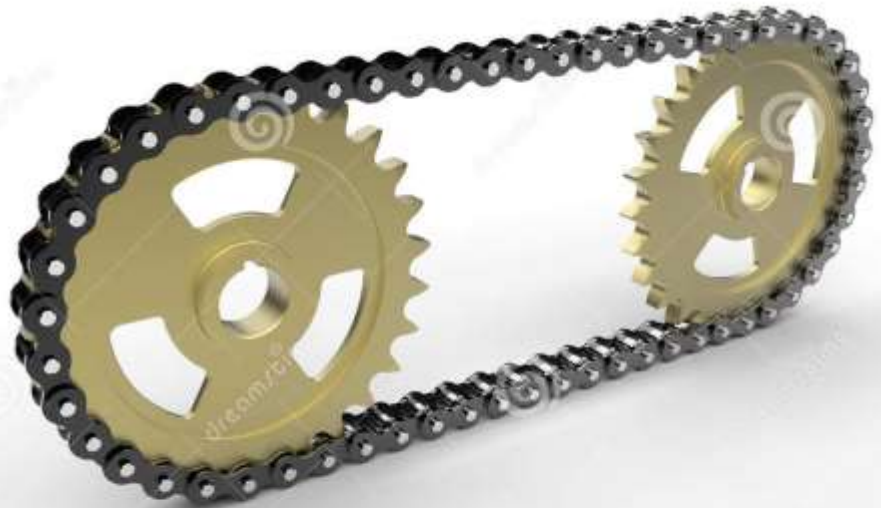
- Transmission of large power over long distances.
- Used in elevators, mine hoist, cranes, conveyors, suspension bridges etc.

Advantages

- Smooth and silent operation.
- Long life
- Withstands shock
- Light weight

Chain drive

- It consists of an endless chain running over special profile toothed wheels called **sprockets**.
- One sprocket – driver, other sprocket – driven.
- Smaller sprocket-**pinion**, Larger sprocket- **wheel**
- Chain is made up of rollers, plates, pins and bushes.
- Parts made of high grade steel



Based on application,

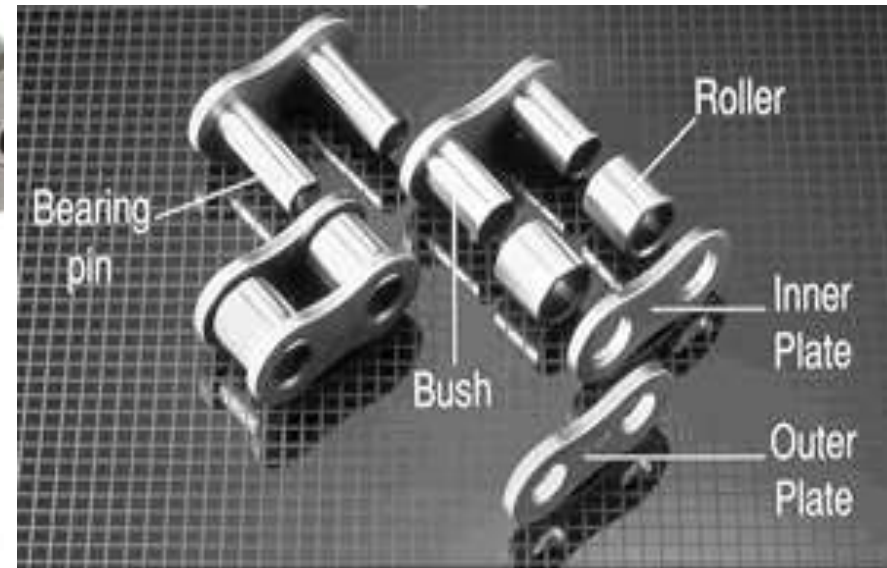
- Power transmission chains
- Hoisting chains
- Pulling chains

Types of chain for power transmission

- Roller chains
- Silent or inverted tooth chain

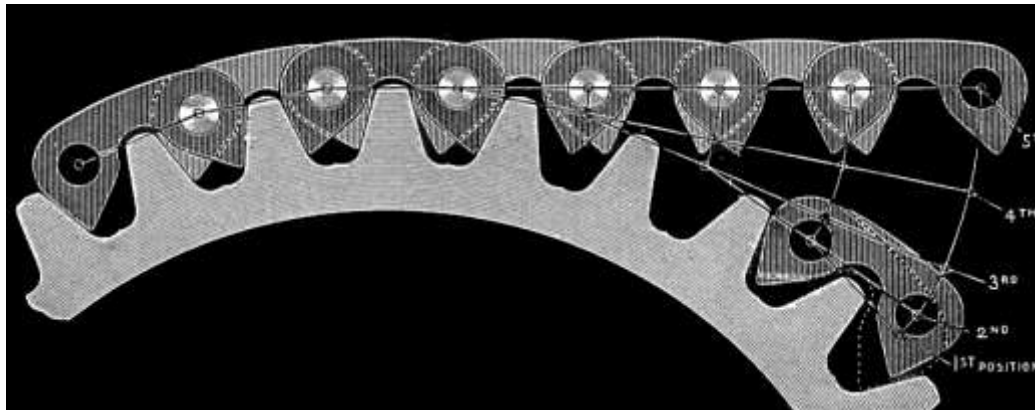
Roller chain

- It consists of rollers, plates, pins and bushes.
- In order to reduce friction, contact surfaces are lubricated.



Silent or inverted tooth chain

- It consists of a no. of flat links, tooth shaped ends corresponding to the sprocket profile.
- Sprockets used are wider than in roller chain drive and have a central groove to keep the chain secure.
- These chains are more complex in design and require careful maintenance.
- Employed to transmit heavier loads.
- It has quieter operation and lesser vibration.



Advantages

- High efficiency of transmission
- Can carry heavy loads
- More durable
- No slip
- No initial tensioning required
- Smaller overall size

Disadvantages

- Careful maintenance and accurate mounting
- Cost high
- Chain stretching

Applications

- Motorcycles
- Automobiles
- Conveyors
- Machine tools
- Agricultural machinery
- Oil-well drilling rigs etc.

Gear Drive

- Gear is a toothed wheel.
- For power transmission, gears are mounted over driver and driven shafts.
- Gears are used to transmit power over short distances and when positive drive is required.
- Idle gears used for longer distance transmission.
- Gears materials:
 - ✓ **Grey iron** – low duty
 - ✓ **Cast steel, alloy steel** – heavy duty
 - ✓ **Nickel, manganese, phosphor bronze** - corrosive environments

Types of gears

1) Spur gears:

- Teeth cut parallel to axis of shaft.
- Transmits power b/w parallel shafts.



2) Helical gears:

- Teeth cut are of helical screw form.
- Helical tooth is inclined at an angle to axis of shaft.



3) Bevel gears:

- Used to connect 2 non-parallel shafts intersecting each other.
- Shafts are generally at right angles to each other.



4) Worm gears:

- Used for power transmission b/w non-intersecting shafts generally at right angles to each other.
- It consists of a worm and a worm wheel.
- Worm: threaded screw, driver
- Worm wheel: toothed wheel, driven



5) Rack and pinion:

- Used to convert rotary motion to rectilinear motion.



Advantages

- High efficiency of transmission
- Long service life
- High reliability
- Constant velocity ratio due to absence of slip.
- Can operate at high speeds.
- Large power can be transmitted.

Disadvantages

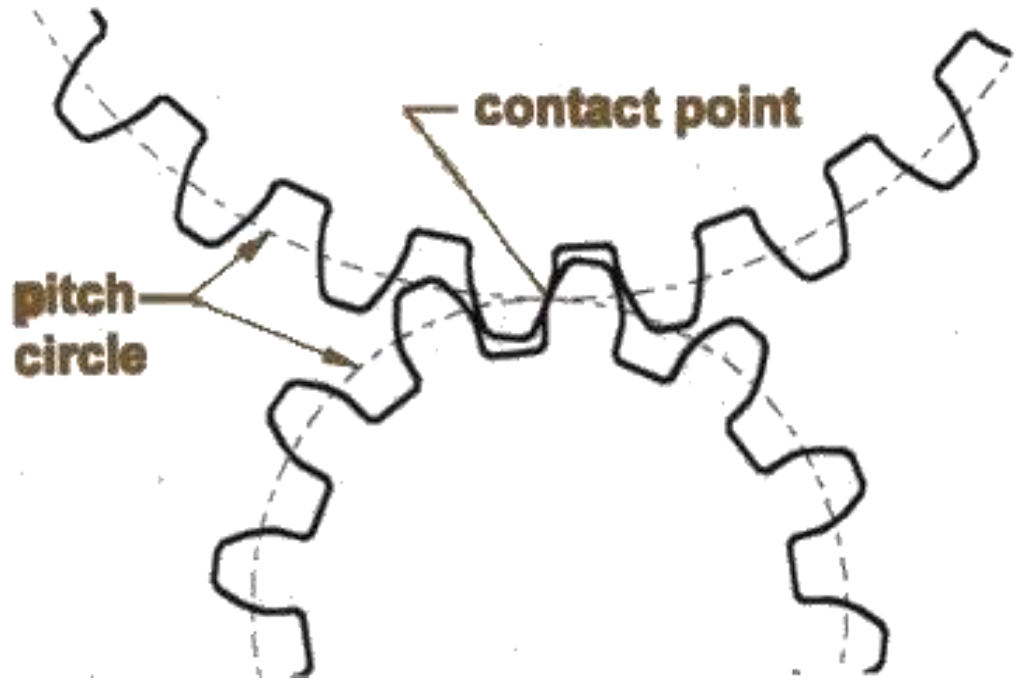
- Damage of a single wheel affects whole setup
- Manufacture of gears requires special tools and equipments.

Applications

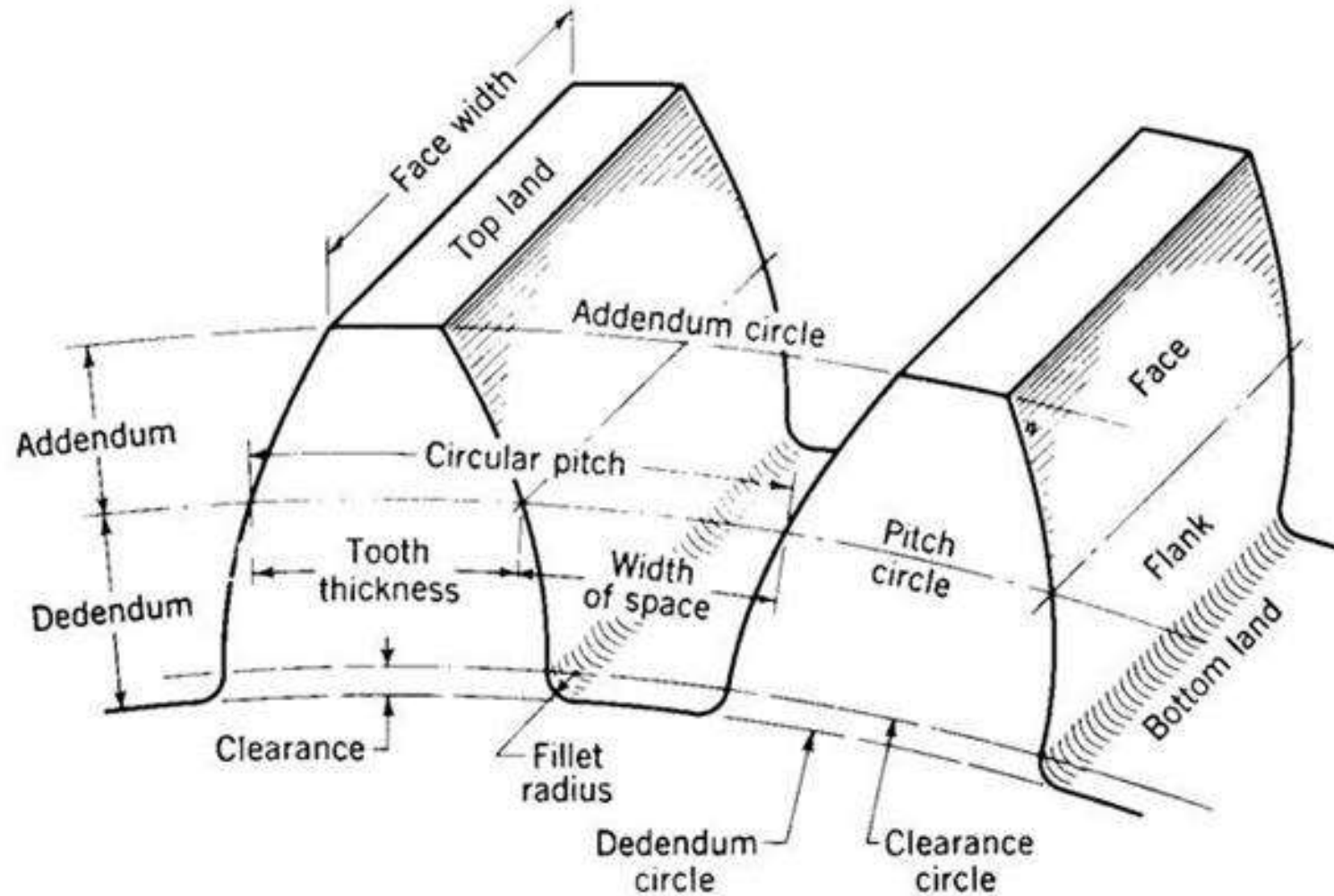
- Used in differentials, transmission gear boxes, planetary drives etc.

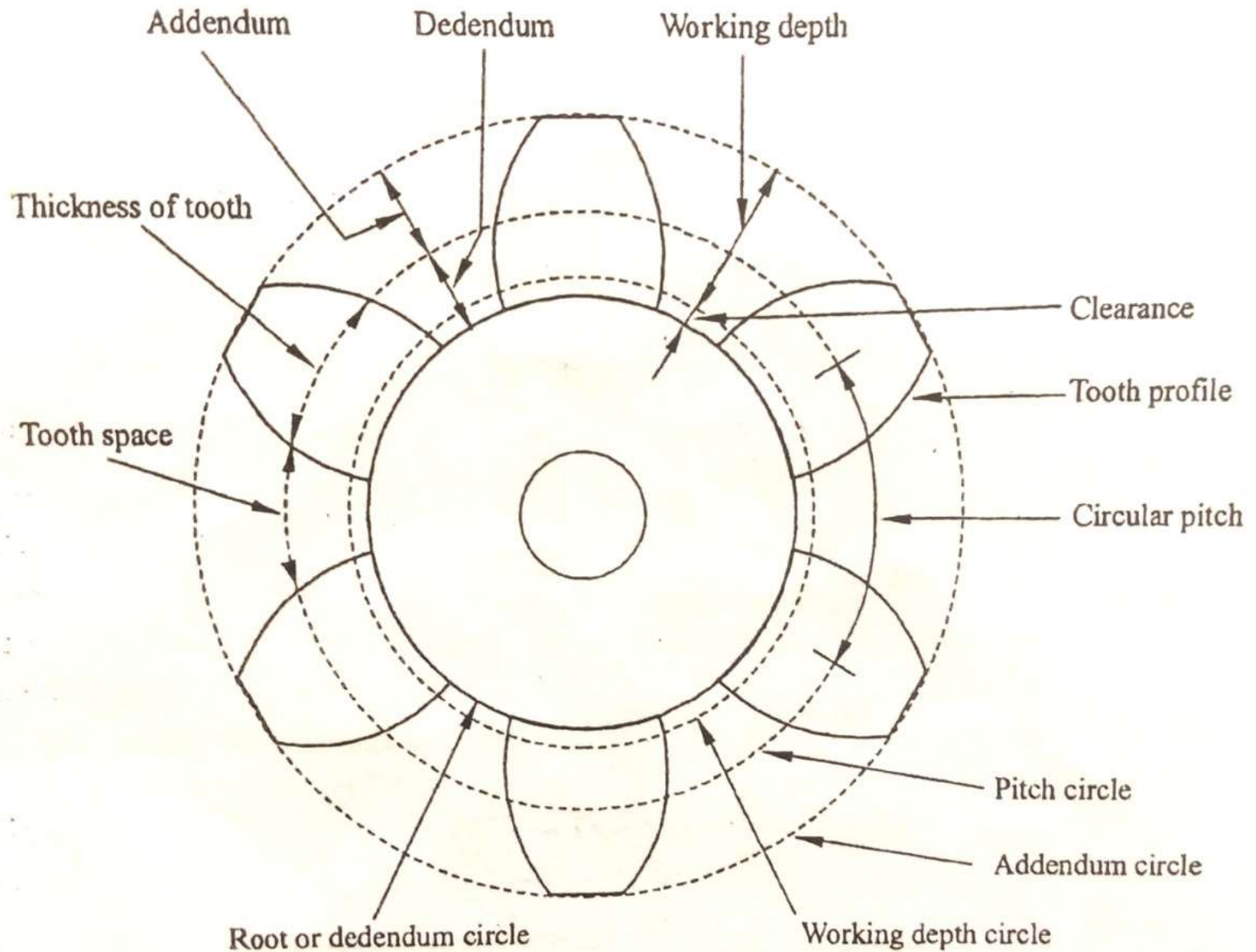
Gear Terminology (or) Nomenclature

- Pitch circle
- Addendum circle
- Addendum
- Dedendum (or) root circle
- Dedendum
- Circular pitch
- Diametrical pitch
- Module
- Top land
- Bottom land
- Face
- Flank
- Face width
- Profile
- Working depth
- Tooth thickness
- Clearance



GEAR NOMENCLATURE





1. GEAR TRAINS

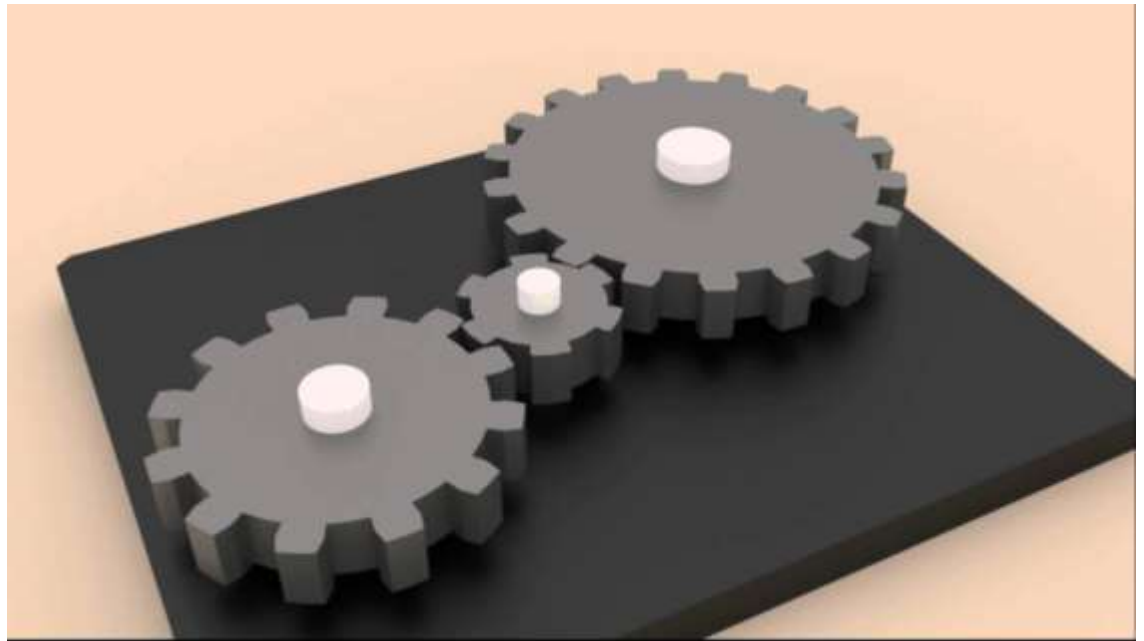
- Any *combination of gear wheels* by which motion is transmitted from one shaft to another shaft.

Types of Gear Trains

1. Simple gear train
2. Compound gear train
3. Reverted gear train
4. Epicyclic gear train

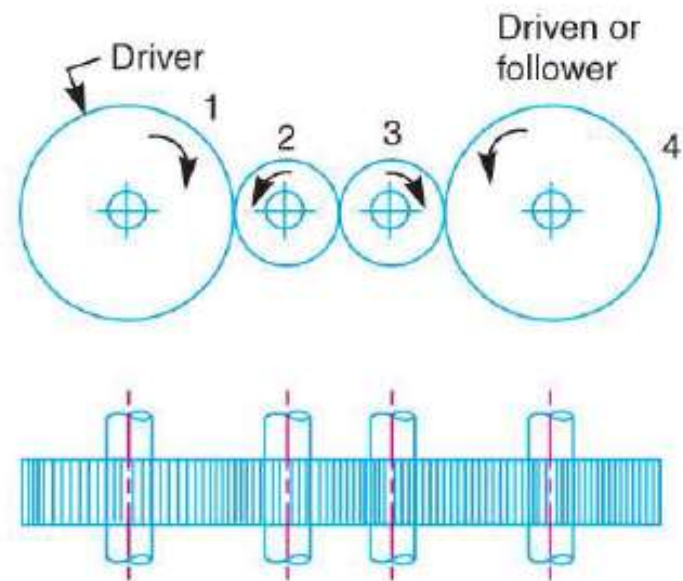
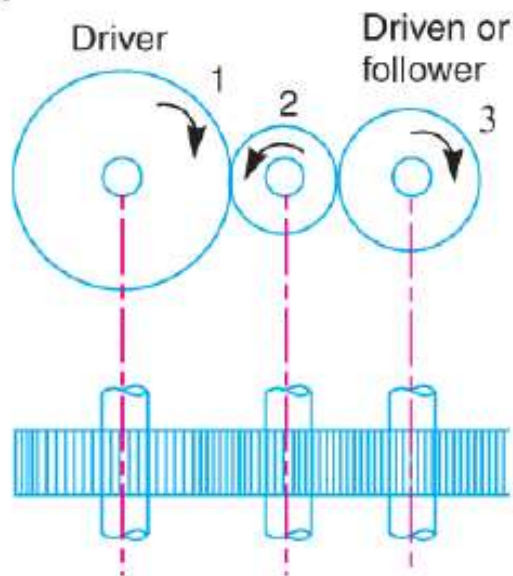
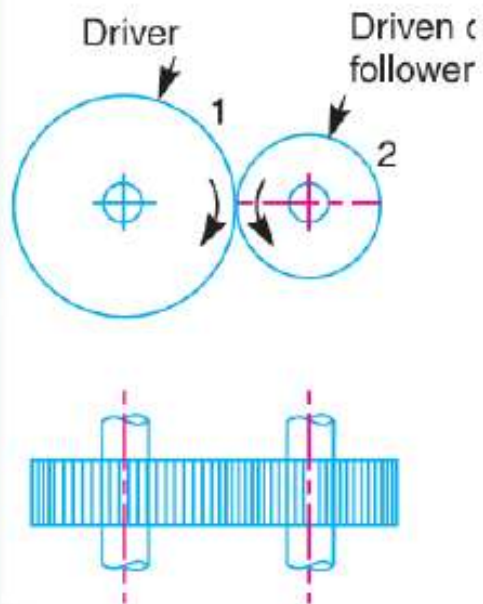
Simple gear train

- Each shaft carries only one gear.
- Intermediate gear acts as both driver and driven.
- Intermediate gears have no effect on velocity ratio - *idle gear*.
- Function of idler gear: *fill gap b/w driver and driven gears.*



Simple Gear train:

When there is only one gear on each shaft, then it is known as simple gear train.



Speed ratio

- ❖ speed ratio (or velocity ratio) of gear train is the ratio of the speed of the driver to the speed of the driven or follower
- ❖ ratio of speeds of any pair of gears in mesh is the inverse of their number of teeth

$$\text{Speed ratio} = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

Train value

- ❖ ratio of the speed of the driven or follower to the speed of the driver is known as train value of the gear train

$$\text{Train value} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$



When the distance between the two gears is large and we need Constant velocity ratio:

❖ The motion from one gear to another, in such a case, may be transmitted by either of the following two methods

1. By providing the large sized gear
2. By providing one or more **intermediate gears**.

❖ The first method (i.e. providing large sized gears) is very inconvenient and uneconomical method

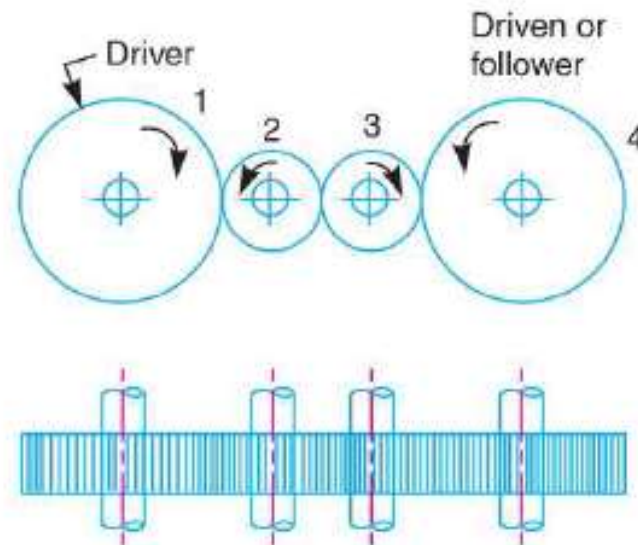
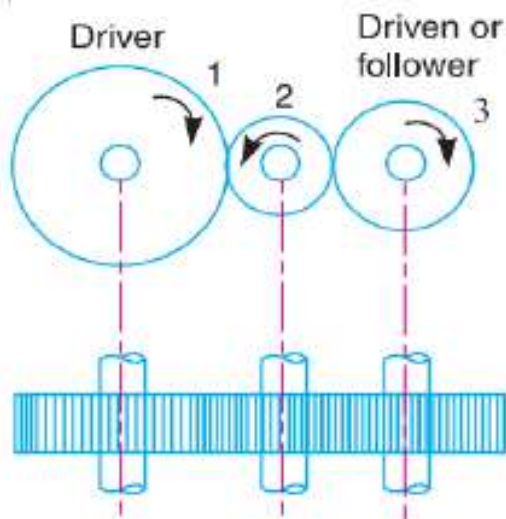
❖ whereas the latter method (i.e. providing one or more intermediate gear) is very convenient and economical.

Intermediate gears

- ❖ intermediate gears are called **idle gears**, as they do not effect the speed ratio or train value of the system.
- 1. To connect gears where a large centre distance is required
- 2. To obtain the desired direction of motion of the driven gear (i.e. clockwise or anticlockwise).

❖ when the number of intermediate gears are odd, the motion of both the gears (i.e. driver and driven or follower) is like.

❖ if the number of intermediate gears are even, the motion of the driven or follower will be in the opposite direction of the driver





The speed ratio of the simple gear train

N_1 = Speed of driver in r.p.m.,

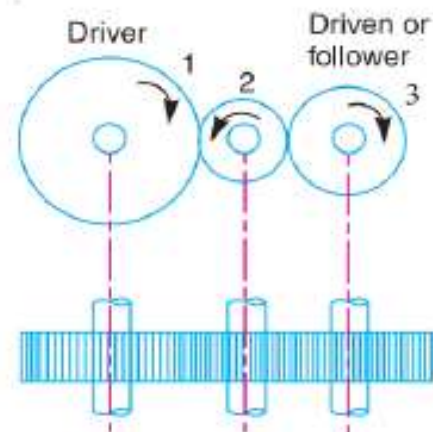
N_2 = Speed of intermediate gear in r.p.m.,

N_3 = Speed of driven or follower in r.p.m.,

T_1 = Number of teeth on driver,

T_2 = Number of teeth on intermediate gear, and

T_3 = Number of teeth on driven or follower.



Since the driving gear 1 is in mesh with the intermediate gear 2,

$$\frac{N_1}{N_2} = \frac{T_2}{T_1}$$

the intermediate gear 2 is in mesh with the driven gear 3

$$\frac{N_2}{N_3} = \frac{T_3}{T_2}$$

The speed ratio of the gear train

$$\frac{N_1}{N_2} \times \frac{N_2}{N_3} = \frac{T_2}{T_1} \times \frac{T_3}{T_2} \quad \text{or} \quad \frac{N_1}{N_3} = \frac{T_3}{T_1}$$



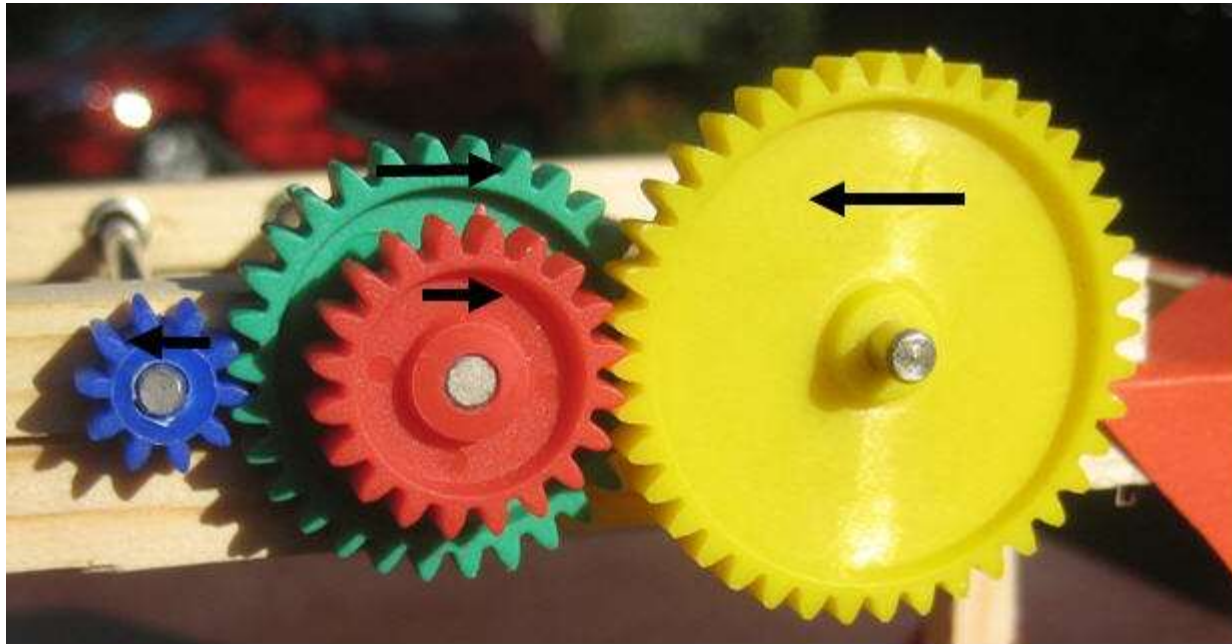
The speed ratio of the simple gear train

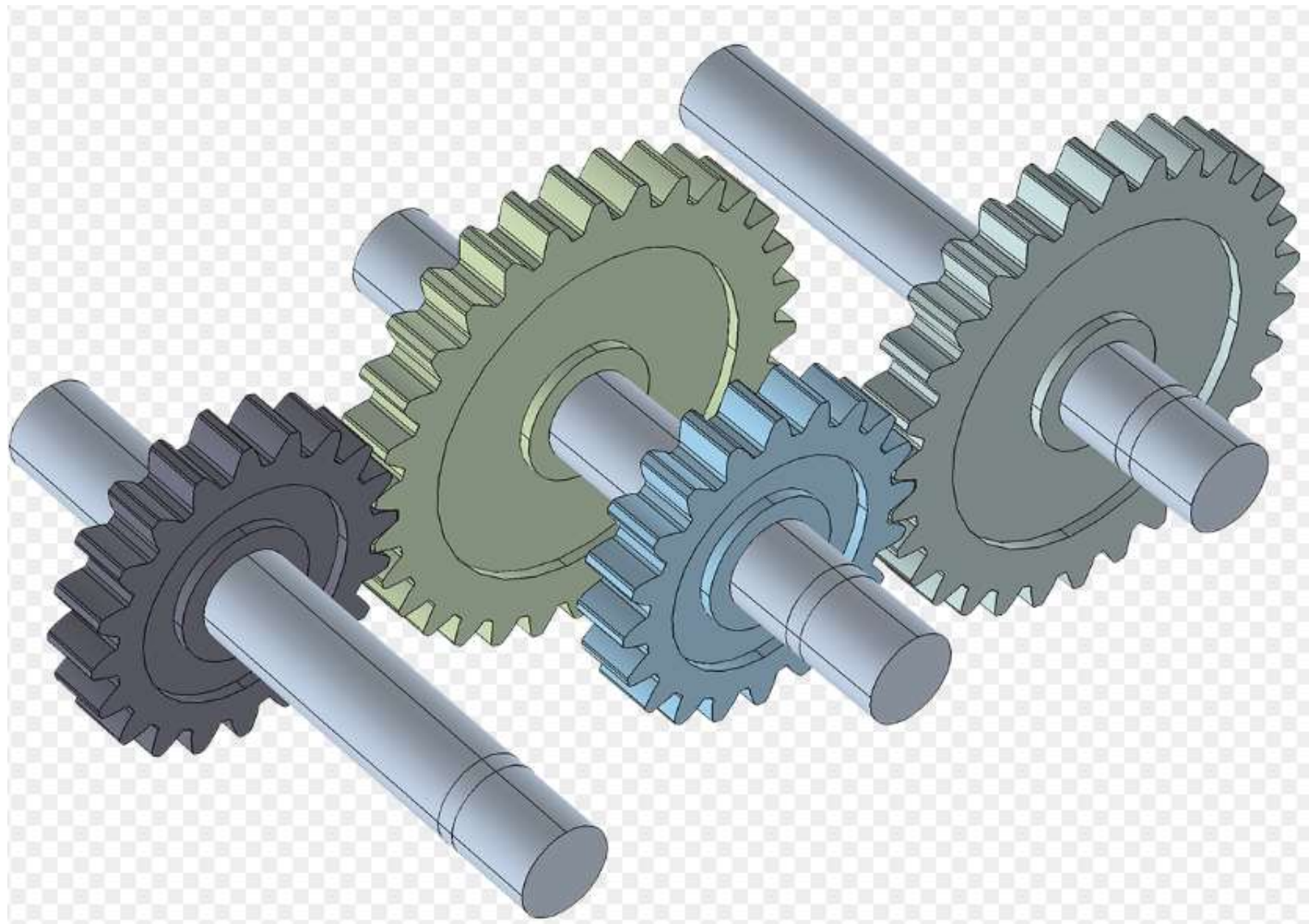
$$\text{Speed ratio} = \frac{\text{Speed of driver}}{\text{Speed of driven}} = \frac{\text{No. of teeth on driven}}{\text{No. of teeth on driver}}$$

$$\text{Train value} = \frac{\text{Speed of driven}}{\text{Speed of driver}} = \frac{\text{No. of teeth on driver}}{\text{No. of teeth on driven}}$$

2. COMPOUND GEAR TRAIN

- A series of gears connected such that two or more gears rotate about the same axis.
- Velocity ratio depends on diameter of driver, driven and intermediate gears.





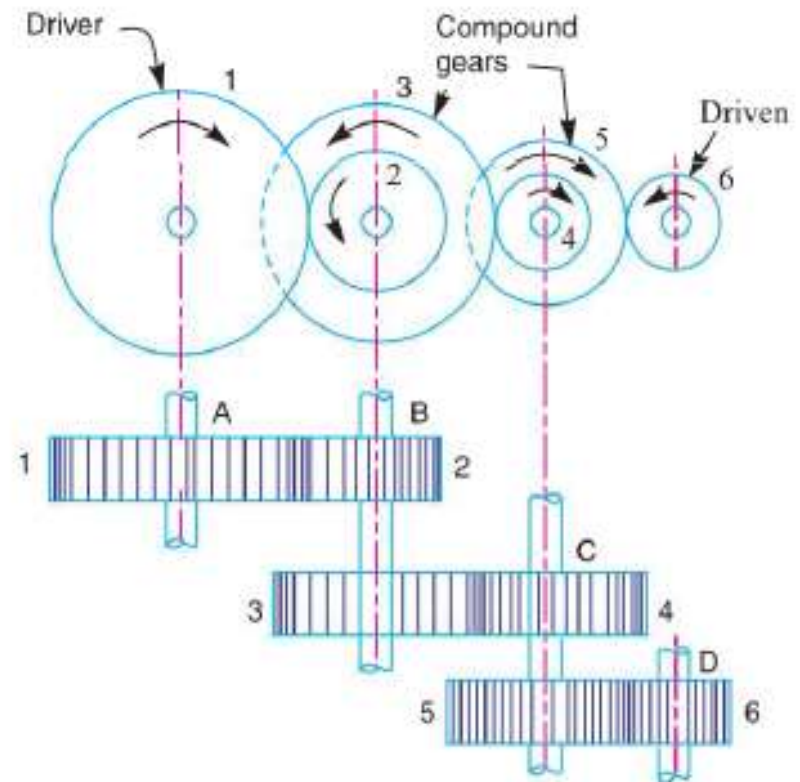
2. Compound Gear Train

- ❖ When there are more than one gear on a shaft, then the gear train is called a compound train of gear.

speed ratio of compound gear train

$$\frac{N_1}{N_2} \times \frac{N_3}{N_4} \times \frac{N_5}{N_6} = \frac{T_2}{T_1} \times \frac{T_4}{T_3} \times \frac{T_6}{T_5}$$

$$* \frac{N_1}{N_6} = \frac{T_2 \times T_4 \times T_6}{T_1 \times T_3 \times T_5}$$

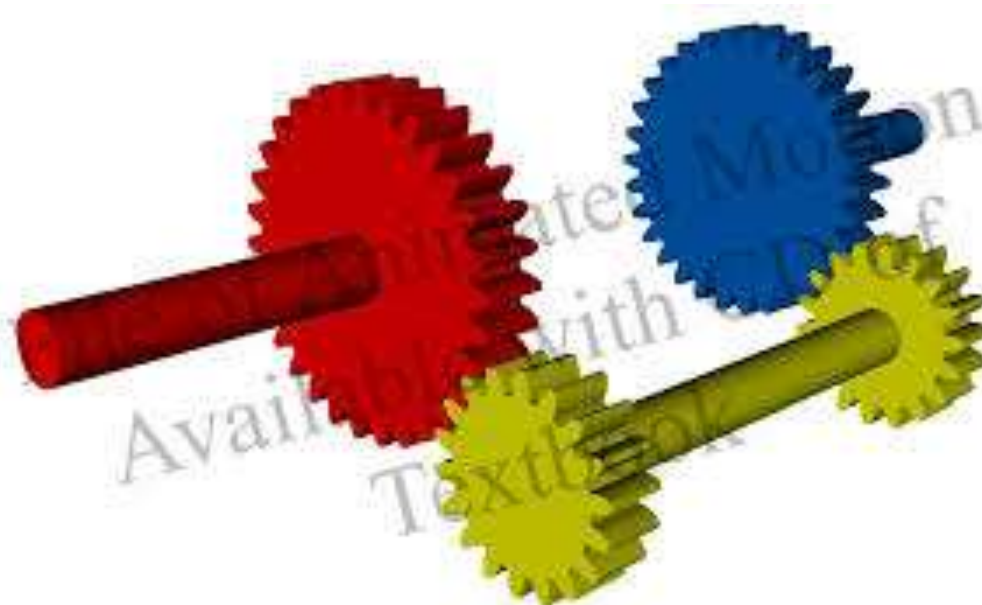


The advantage of a compound train over a simple gear train

- ❖ much larger speed reduction from the first shaft to the last shaft can be obtained with small gears.
- ❖ If a simple gear train is used to give a large speed reduction, the last gear has to be very large.
- ❖ Usually for a speed reduction in excess of 7 to 1, a simple train is not used and a compound train or worm gearing is employed.

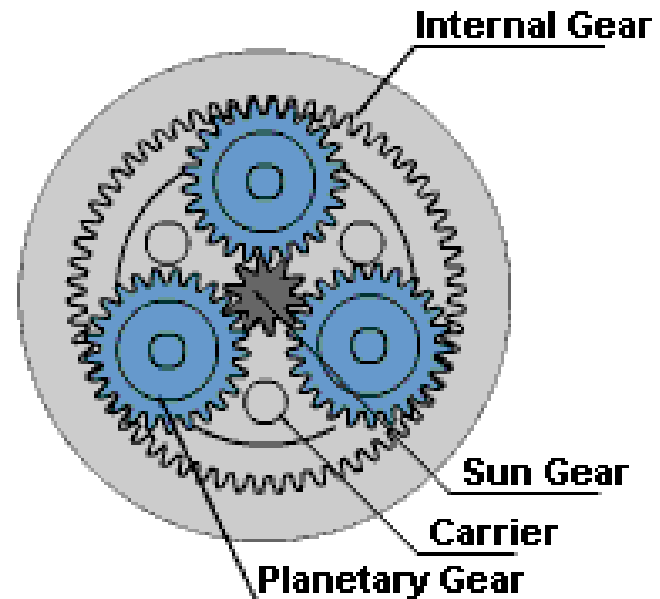
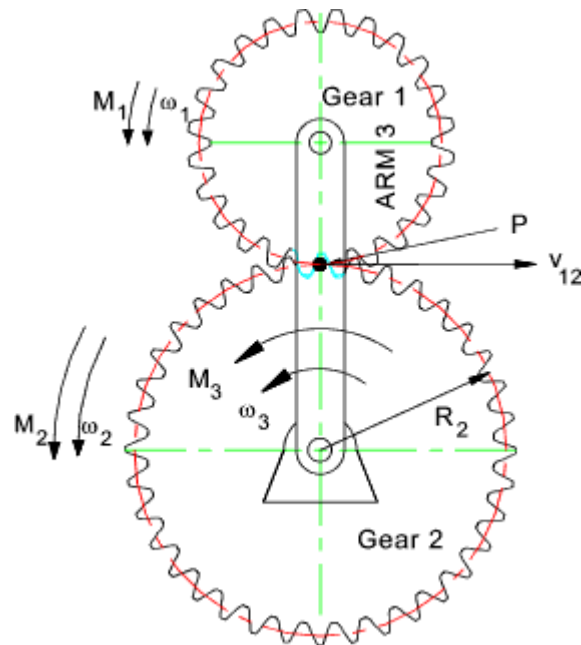
3. REVERTED GEAR TRAIN

- When axis of first gear (i.e. first driver) and the last gear (i.e. last driven or follower) are **co-axial**, it is a reverted gear train.
- In a reverted gear train **direction of rotation** of first and last gear are **same**



4. Epicyclic gear train (Planetary gear train)

- For other gear trains, axis of gears remain fixed.
- Here, there *is relative motion b/w the axis of 2 or more wheels.*
- Useful for transmitting *very high velocity ratios* with gears of moderate size in lesser space.



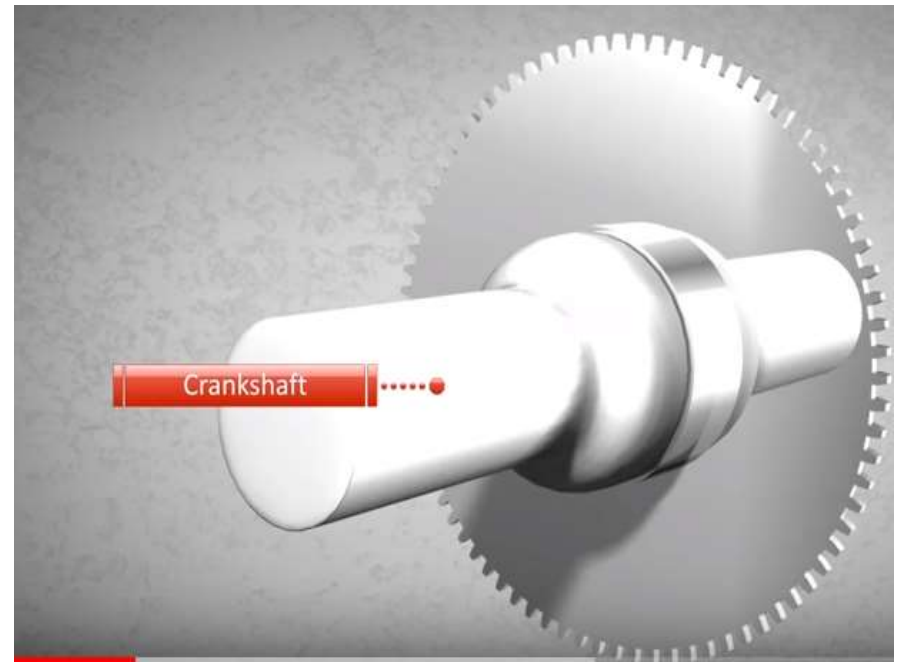
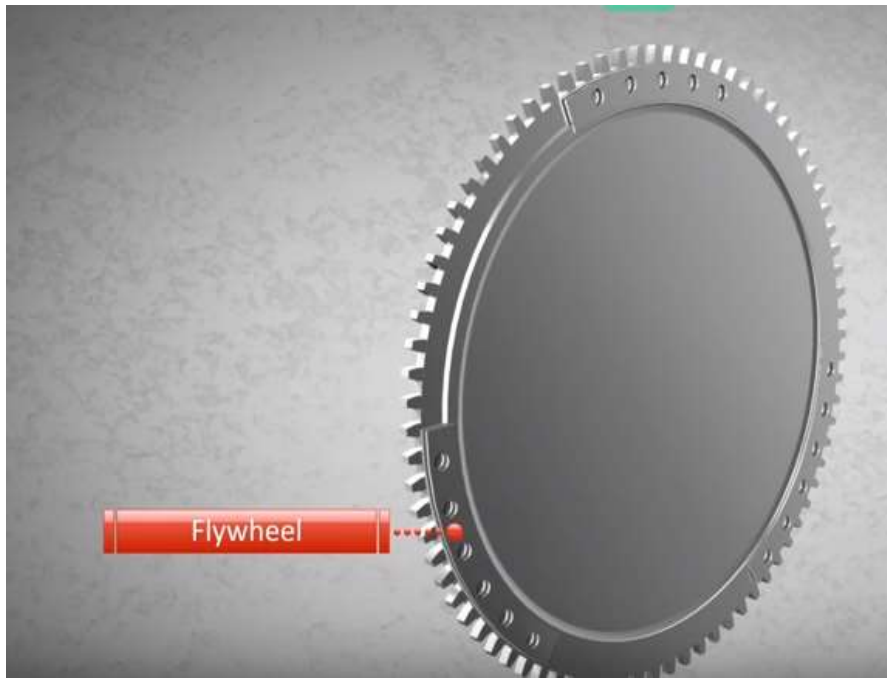
Applications

- Electrical machines
- Differential gears
- Gear boxes of automobiles etc.

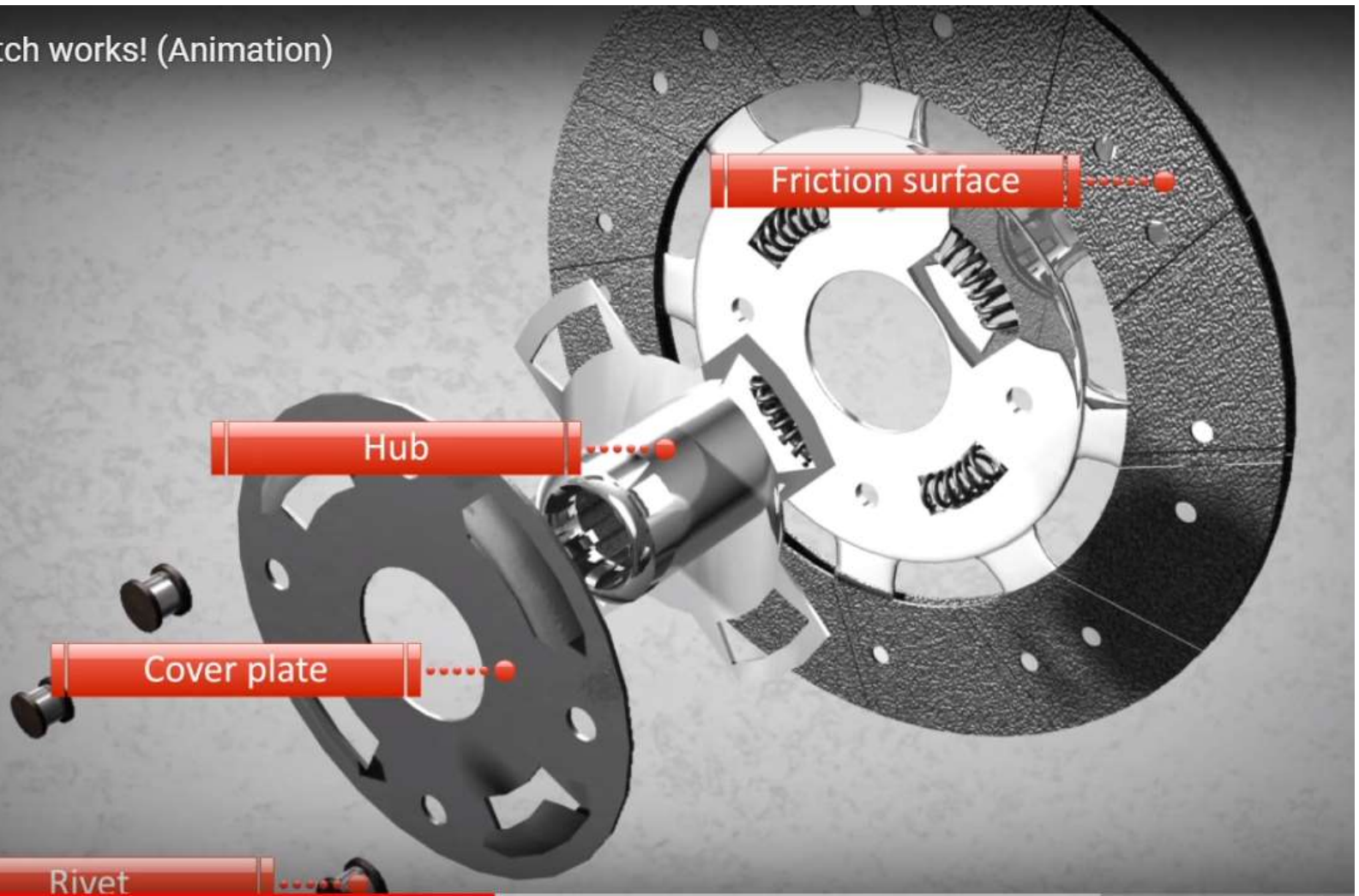
SINGLE PLATE CLUTCH

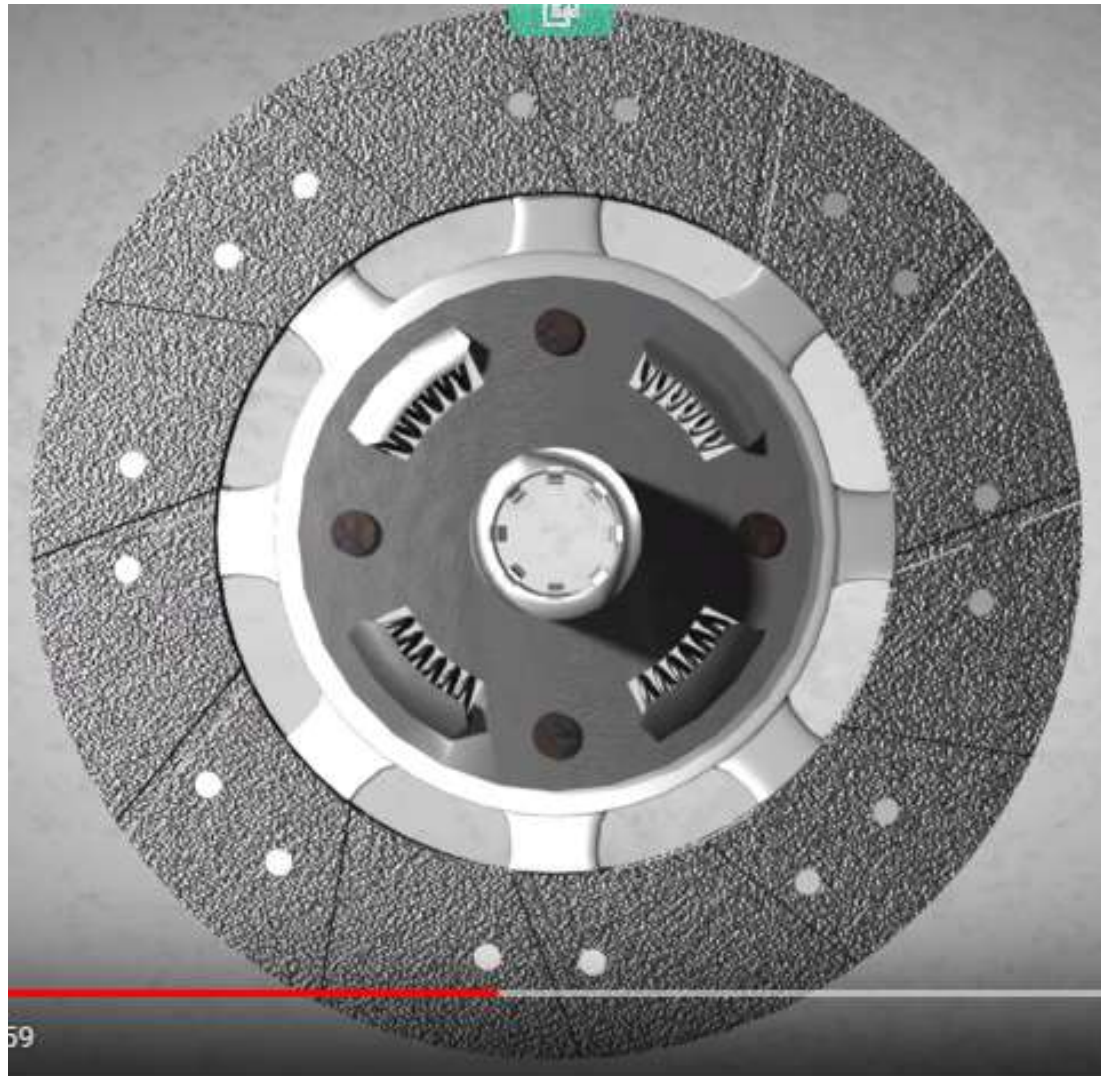
Parts

1. Flywheel which is attached to crank shaft (driver shaft)
2. Clutch plate with friction lining on both sides
3. Hub
4. Pressure plate
5. Spring
6. Shaft to gear box (driven shaft)

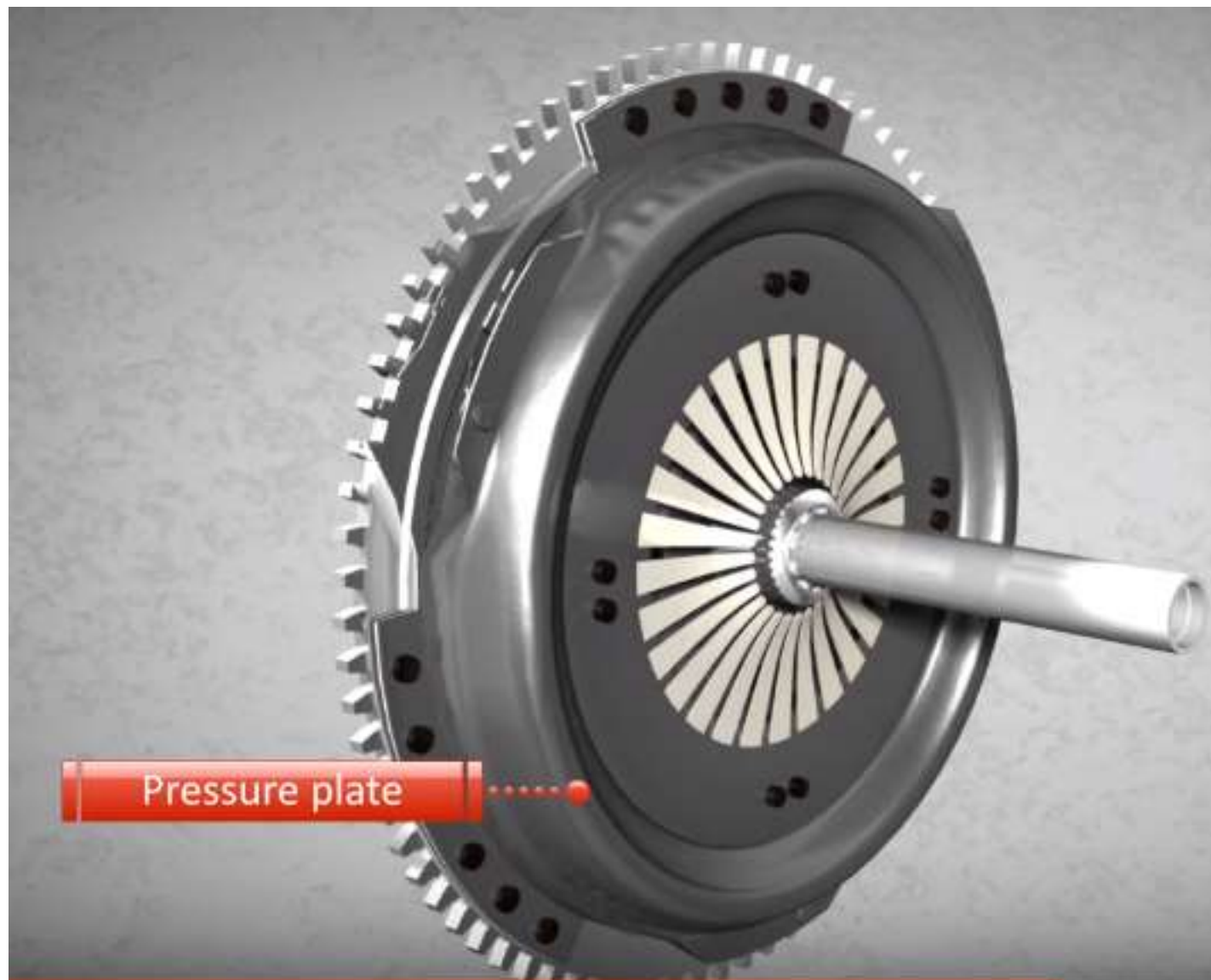


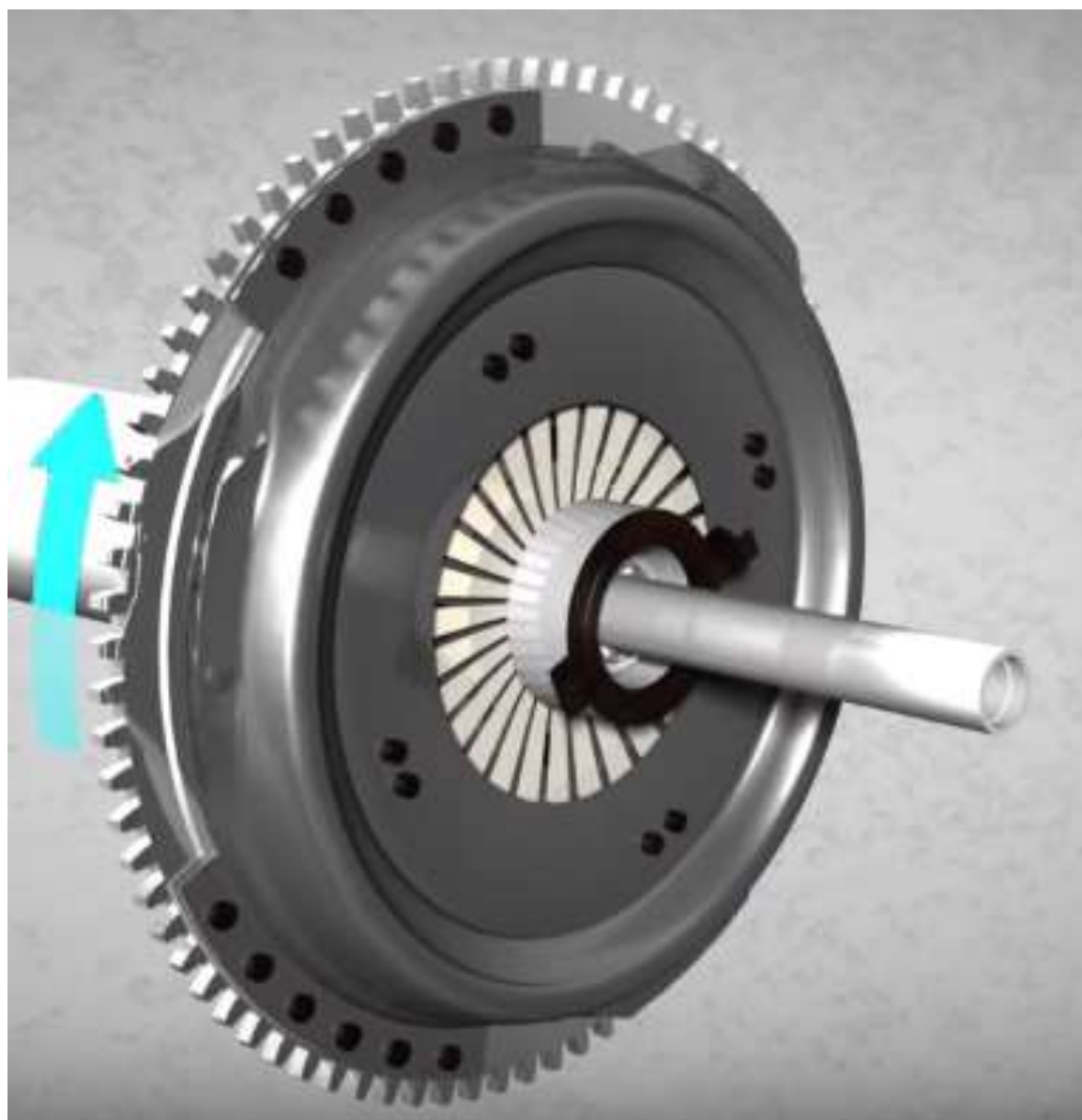
How a clutch works! (Animation)



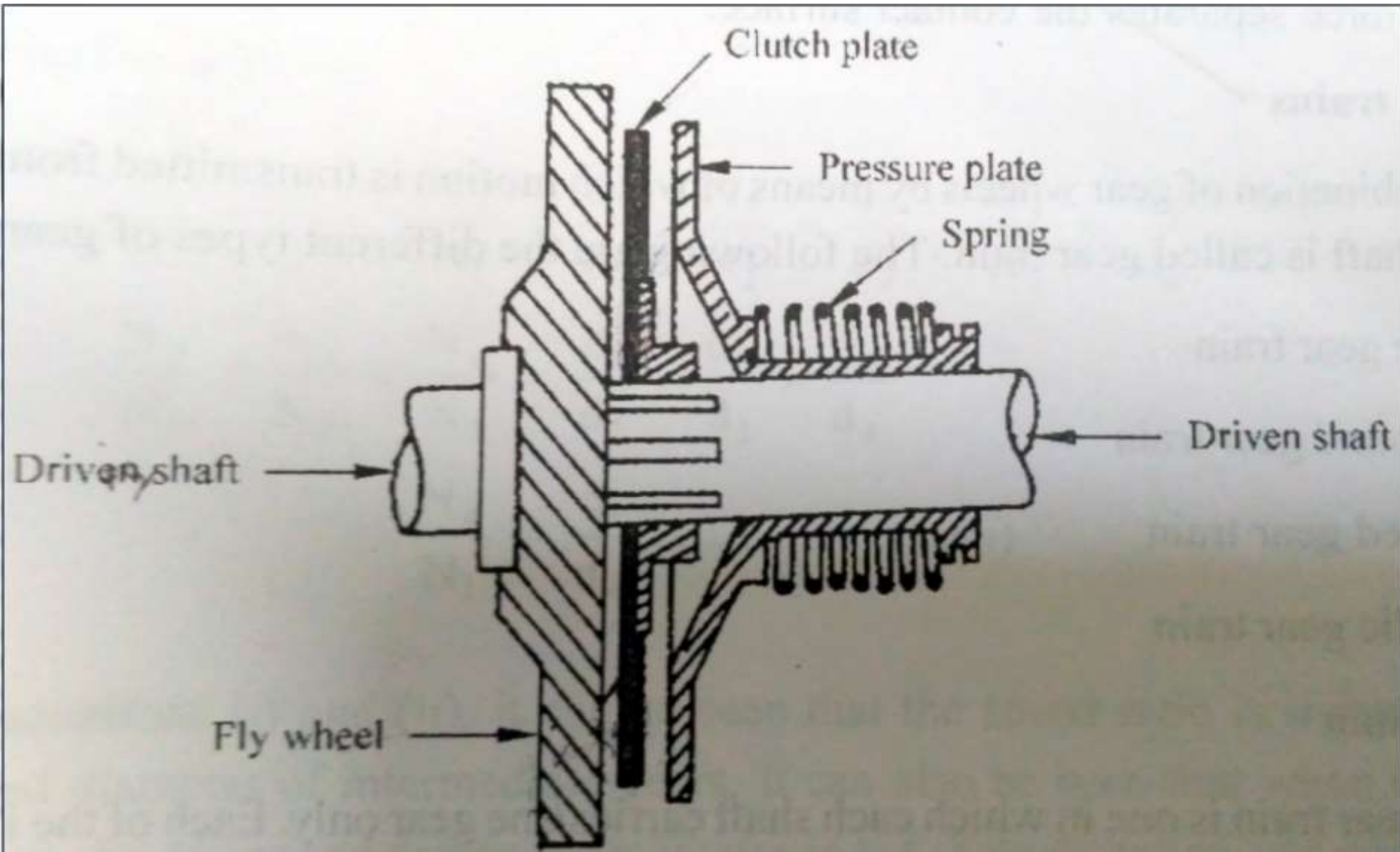


Clutch Plate





Single Plate Clutch



- Transmit power by *friction*- Friction Clutch
- *Clutch plate* made of steel having ceramic frictional lining on both sides
- Clutch plate is attached to *hub* which rotates with driven shaft
- When clutch is engaged
 - Spring loaded *pressure plate* presses clutch plate against flywheel
 - Friction b/w lining of clutch plate and flywheel on one side & Friction b/w lining of clutch plate and pressure plate on other side cause clutch plate and driven shaft rotate when clutch is engaged

- When clutch is disengaged

- *Compression of spring* pulls back the pressure plate
- Contact between flywheel and clutch plate breaks
- Fly wheel will rotate freely
- *No power is transmitted* from engine to gear box

CONE CLUTCH

